Bernoulli Tasks 1 and 2
Revised framework and edit instructions

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Apply the changes below to the **Bernoulli sections only**. Both tasks compare a binary model decision with annotator B's observed binary label. The tasks deliberately use different references. This memo specifies the revised targets, assumptions, tests, and required manuscript and experiment edits.

Task	Compared outputs	References and loss
Task 1	Binary model decision and observed label of B	Observed labels of A and C; 0/1 error differences.
Task 2	Binary model decision and observed label of B	Mean reference-model probability and latent probability; squared-loss differences.

Common Bernoulli model and notation

Keep the probability model and logit-space annotation bias. Expectations over items are denoted by E or E_X; annotator identities are held fixed unless an identity expectation is displayed. Distinct annotators' evaluation labels are conditionally independent given the item and their identities.

$$Y_r \mid X, \xi_r \sim \text{Bernoulli}(\mu_r(X)), \qquad \text{logit}\, \mu_r(X) = \text{logit}\, \mu^*(X) + b_r(X). \tag{1}$$

Train the probability model on A's labels using BCE. Let t denote the threshold after it has been fixed, either in advance or using independent validation data. Use the same tie convention throughout: a score equal to t predicts 1.

$$\hat{Y}_A(x) = \mathbf{1}\{\hat{\mu}_A(x) \geq t\}, \qquad \hat{\mu}_R(x) = \frac{1}{K} \sum_{r \in R} \hat{\mu}_r(x). \tag{2}$$

The conventional equal-cost threshold is t = 1/2. Validation-based selection is allowed under the conditions on page 5. Reference models continue to estimate probabilities; do not threshold their outputs.

Meaning of the Task 2 latent reference

Task 2 measures the loss of the two binary outputs relative to the latent probability function itself. It does not introduce a generated latent binary label. This avoids extra label-generation noise on the reference side and keeps the chosen latent estimand explicit. The randomness in B's observed output remains part of the comparison.

Source documents and scope

[S1] Anno_bias_theory_modified(8).pdf: use Section 4 for the original Task 1 comparison and paired test.
[S2] Bernoulli_version_Sept(4).pdf: use Sections 1-6 for the Bernoulli/logit Task 2 model, calibration, bound, and certification rule.
[S3] Supplied screenshot titled "Assumptions for binary labels": the three displayed logit assumptions are addressed on page 5.

1. Task 1: revised target and paired test

The competitors are the fixed binary model decision and B's observed label. The two references are A's and C's **held-out evaluation labels**. They are not the training labels used to fit the model.

$$\begin{aligned}\Delta_R &:= \mathbb{E}\Big[\mathbf{1}\{\hat{Y}_A \neq Y_R\} - \mathbf{1}\{Y_B \neq Y_R\}\Big] \\ &= \mathbb{E}\Big[(\hat{Y}_A - Y_R)^2 - (Y_B - Y_R)^2\Big], \qquad R \in \{A, C\}.\end{aligned} \tag{3}$$

A positive comparison means that the model is worse than B under that reference. The equality between the indicator and squared-loss expressions holds because both outputs and the reference are binary.

$$\begin{aligned}S_{A,C} &:= \Delta_C - \Delta_A \\ &= 2\mathbb{E}[(Y_A - Y_C)(\hat{Y}_A - Y_B)].\end{aligned} \tag{4}$$

Thus $S_{A,C} > 0$ means that reference A makes the model look better relative to B than reference C does. This is a change in the model-human comparison; it does not by itself assert a reversal of the comparison's sign.

Observable statistic and standard error

For m independent held-out evaluation items, use

$$\hat{S}_{A,C} = \frac{2}{m}\sum_{j=1}^{m}(y_{A,j} - y_{C,j})\{\hat{Y}_A(x_j) - y_{B,j}\}. \tag{5}$$

Compute the standard error as the sample standard deviation of the **full paired summands** $2(y_{A,j} - y_{C,j})(\hat{Y}_A(x_j) - y_{B,j})$, divided by √m. Do not treat the two reference comparisons as independent samples.

$$H_0 : S_{A,C} \leq 0, \qquad H_1 : S_{A,C} > 0, \qquad \text{reject if } \frac{\hat{S}_{A,C}}{\widehat{\mathrm{se}}(\hat{S}_{A,C})} > t_{m-1,1-\alpha}. \tag{6}$$

Validity statement to retain

Condition on the trained model and the frozen threshold-selection data. Under independent held-out evaluation items and finite, nonzero variance of the paired summand, the one-sided paired test is asymptotically level α. The summands are bounded, so finite moments are automatic; nondegenerate variance is still required.

The original paired-CLT argument in [S1, Section 4.2] is unchanged after replacing the model probability by the binary decision. The Student-t cutoff is used as an asymptotic cutoff; binary summands do not imply an exact finite-sample t distribution.

Required calculation changes

- Replace the model probability in both reference losses and in the paired statistic by the frozen binary decision.
- Recompute effect estimates, paired standard errors, p-values, and rejection rates. Keep the observed labels of A, B, and C.

2. Task 1: replace the population explanation

The conditional-independence assumption removes the evaluation-label noise from the expectation. The correct Bernoulli decomposition is

$$\frac{1}{2}S_{A,C} = \mathbb{E}_X[(\mu_A - \mu_C)(\hat{Y}_A - \mu_B)]. \tag{7}$$

In this expression, each probability is obtained from the logit model in (1). Do not insert logit bias directly into an additive probability decomposition.

Ideal learning with a common fixed threshold

Suppose the model learns the annotator probability exactly and t is common and fixed for the annotator-population calculation. For iid annotator identities A, B, C,

$$\mathbb{E}_{A,B,C}[S_{A,C}] = 2\mathbb{E}_X[\mathrm{Cov}_A(\mu_A(X), \mathbf{1}\{\mu_A(X) \geq t\})] \geq 0. \tag{8}$$

To obtain this expression, expand (7), average over the independent identities B and C, and cancel their mean terms. The remaining covariance is nonnegative because the threshold indicator is nondecreasing in the probability. Itemwise centering of logit bias does not need to imply centering on the probability scale for this calculation.

Replace the old variance formula. The right-hand side is not the old $2E_X[\tau_X{}^2]$. Strict positivity is not automatic. If, for each item, all annotator probabilities lie on the same side of t, the covariance can be zero despite variation in their probabilities.

Finite probability-estimation error

Keep $h_A = \hat{\mu}_A - \mu_A$ as a probability-estimation error. Define $S^0_{A,C}$ only for this comparison: it is (4) with the model decision replaced by $1\{\mu_A \geq t\}$, using the same threshold.

$$\begin{aligned} S_{A,C} - S^0_{A,C} &= 2\mathbb{E}_X[(\mu_A - \mu_C)\{\hat{Y}_A - \mathbf{1}\{\mu_A \geq t\}\}], \\ |S_{A,C} - S^0_{A,C}| &\leq 2P_X(\hat{Y}_A \neq \mathbf{1}\{\mu_A \geq t\}) \\ &\leq 2P_X(|\mu_A - t| \leq \eta) + \frac{2\|h_A\|^2_{L^2(P_X)}}{\eta^2}, \qquad \eta > 0. \end{aligned} \tag{9}$$

The last inequality follows because a disagreement requires the estimation error to reach the distance to the threshold. A convergence rate needs control of probability mass near t as well as a probability-estimation error bound. Do not copy the old linear h_A remainder or claim that thresholding preserves its rate. Any BCE estimation rate must be justified for BCE; [S1]'s MSE-ERM bound does not automatically become a BCE bound.

When thresholds are selected separately for each annotator

The paired test on page 2 remains valid under independent validation and evaluation. However, (8) with its nonnegative sign is a **common fixed-threshold** population statement. If the selected threshold depends on A, the resulting decision need not be monotone in μ_A across identities. Do not claim automatic nonnegativity for that adaptive population comparison. Use (7) and the held-out test for the fitted classifier.

No margin condition is needed for the paired test itself. The near-threshold condition in (9) belongs only to the learning-error analysis.

3. Task 2: latent target and reference distortion

Keep the reference set disjoint from A and B. The reference is the average of the fitted reference probabilities, not a majority vote or an average of thresholded reference decisions. Condition on the fitted functions when taking evaluation expectations.

$$\begin{aligned} \Delta^* &:= \mathbb{E}[(\hat{Y}_A - \mu^*)^2 - (Y_B - \mu^*)^2] \\ &= \mathbb{E}[(\hat{Y}_A - Y_B)(1 - 2\mu^*)], \\ \Delta_{\hat{R}} &:= \mathbb{E}[(\hat{Y}_A - \hat{\mu}_R)^2 - (Y_B - \hat{\mu}_R)^2] \\ &= \mathbb{E}[(\hat{Y}_A - Y_B)(1 - 2\hat{\mu}_R)]. \end{aligned} \tag{10}$$

These are squared-loss comparisons to probability functions. Keep that description in the manuscript. Do not describe a fractional reference as an observed binary ground-truth label, and do not replace the latent target by its thresholded value.

$$\begin{aligned} g_{\hat{R}} &:= \mathbb{E}[(\hat{\mu}_R - \mu^*)(\hat{Y}_A - Y_B)], \\ \Delta_{\hat{R}} - \Delta^* &= -2g_{\hat{R}}, \\ \Delta_{\hat{R}}\Delta^* > 0 \quad &\Longleftrightarrow \quad \Delta_{\hat{R}}(\Delta_{\hat{R}} + 2g_{\hat{R}}) > 0. \end{aligned} \tag{11}$$

Replace the probability-scale error condition

$$h_r := \hat{\mu}_r - \mu_r, \qquad h_R := \frac{1}{K}\sum_{r \in R} h_r, \qquad \mathbb{E}[h_R(X)\{\hat{Y}_A(X) - Y_B(X)\}] = 0. \tag{12}$$

Equation (12) is the moment condition needed by the proof. If retaining the original two-assumption presentation in [S2, Section 2], keep $E[h_R] = 0$ and replace its independence condition by

$$h_R(X) \perp\!\!\!\perp \{\hat{Y}_A(X) - Y_B(X)\}. \tag{13}$$

The old independence or orthogonality condition involving $\hat{\mu}_A - Y_B$ does not automatically imply (12) or (13). Centered logit error also does not imply (12). Separate training and evaluation are needed for the sampling analysis but do not, by themselves, establish this orthogonality condition.

Reference contributions used in the bound

$$g_r := \mathbb{E}_X[(\mu_r - \mu^*)(\hat{Y}_A - \mu_B)], \qquad g_{\hat{R}} = \frac{1}{K}\sum_{r \in R} g_r. \tag{14}$$

The equality follows from (12) and $E[Y_B \mid X] = \mu_B$. For concentration across reference identities, the frozen model decision must be independent of the sampled reference identities. This is the reason not to select its threshold using the same reference panel.

The fundamental squared-loss identity (11) always holds. The extra error condition is needed specifically to reduce its distortion to the reference contributions controlled by the original logit analysis.

4. Assumptions and threshold-selection protocol

The three assumptions in screenshot [S3] concern reference probabilities and reference logit biases. Their mathematical forms remain unchanged when only the compared model output is thresholded.

The three displayed logit assumptions: retain

$$\mathbb{E}_X[\operatorname{logit} \hat{\mu}_r(X) - \operatorname{logit} \mu_r(X)] = 0. \tag{15}$$

Centered logit error. This concerns the reference model, not A's threshold. Exact centering is a sufficient first-stage bias condition for logit contrasts; it is separate from probability-scale centering and from (12).

$$C_r := \mathbb{E}_X[b_r(X)], \qquad C_r \overset{\text{iid}}{\sim} N(0, \tau^2). \tag{16}$$

Gaussian global logit offsets. Retain the normal model on C_r. Do not move that Gaussian assumption to g_r; the Bernoulli proof uses bounded-contribution concentration.

$$\mathbb{E}_r \operatorname{Var}_X(b_r(X)) \le (\kappa - 1)\tau^2, \qquad \kappa \ge 1. \tag{17}$$

Bounded item-dependent variation. Retain the same sensitivity parameter κ. Together with (16), it gives $E_X E_r b_r^2 \le \kappa\tau^2$. Also retain the original itemwise centering $E_r b_r(x) = 0$; it is needed to cancel the first-order sigmoid term.

Choose and freeze the threshold before final evaluation

- Fit the probability model, choose the threshold by a documented criterion on independent validation data, then freeze both. Specify how ties between candidate thresholds are resolved.
- The final evaluation set D_1 must not be used to choose or revise the threshold. Condition on the training and threshold-selection data for the held-out paired analysis.
- For Task 2, do not use labels or predictions from the same reference panel R to select the threshold. Separate validation items alone do not remove dependence on those reference identities.
- A straightforward design is: use A-labeled validation data, independent of D_1 and the sampled panel. Keep validation separate from D_2 as well for a simple implementation. Independence of D_1 and D_2 from each other is not required by the union-bound proof.

Searching multiple thresholds on this independent validation set needs no multiplicity correction for the single frozen classifier's final held-out test. Selecting a threshold after inspecting final test statistics or certification outcomes is a different procedure and is not covered here.

For Task 2, the bound still uses $|\hat{Y}_A - \mu_B| \le 1$. Thresholding changes neither this bound nor the logit calibration formula. For Task 1, keep the distinction between held-out test validity and the fixed-threshold population claim on page 3.

5. Task 2: retain calibration and the bias bound

Calibration from reference probabilities on D_2

Keep $K > 1$ fixed in the stated asymptotic calibration. With reference 1 as baseline, define the signed average contrasts and their vector:

$$w_r := \frac{1}{|D_2|}\sum_{i\in D_2}\{\operatorname{logit}\hat{\mu}_r(x_i) - \operatorname{logit}\hat{\mu}_1(x_i)\}, \qquad w = (w_2,\ldots,w_K)^\top. \tag{18}$$

Let $\hat{\Omega}$ be the sample covariance of the **itemwise contrast vector** inside these averages. For a candidate offset scale $u \geq 0$, use

$$\begin{aligned} \hat{V}_\tau(u) &= u^2(I + \mathbf{1}\mathbf{1}^\top) + \frac{\hat{\Omega}}{|D_2|}, \qquad \hat{Q}_\tau(u) = w^\top \hat{V}_\tau(u)^{-1} w, \\ U_{\tau,\gamma}(|D_2|) &= \sup\{u \geq 0 : \hat{Q}_\tau(u) \geq q_{\gamma,K-1}\}. \end{aligned} \tag{19}$$

Here I and the all-ones vector have dimension $K-1$, and $q_{\gamma,K-1}$ is the lower γ quantile of χ^2_{K-1}. Use zero if the inversion set is empty. Retain the coverage requirement $P\{\tau \leq U_{\tau,\gamma}\} \geq 1 - \gamma + o(1)$.

Keep the existing regularity conditions: reference outputs strictly between 0 and 1; out-of-sample evaluation; fixed-dimensional CLT and covariance consistency; nonsingularity where inverted; and inclusion of reference-training uncertainty if it is nonnegligible at the calibration scale. Clipping must satisfy the same first-stage accuracy condition. The contrast-bias formulation in [S2, Eq. (6)] is

$$\mathbb{E}_X[\operatorname{logit}\hat{\mu}_r - \operatorname{logit}\hat{\mu}_1] = C_r - C_1 + o_p(|D_2|^{-1/2}). \tag{20}$$

The exact centered-logit assumption (15) implies the bias requirement in (20), but does not replace the CLT or covariance conditions. The thresholded decision is absent from (18)-(20).

Why the same distortion bound survives

Fix A, B, and the classifier independently of the sampled reference identities. With the revised g_r in (14), sigmoid curvature, Lipschitz continuity, and bounded output differences give

$$\begin{aligned} |\mathbb{E}_r g_r| &\leq \frac{\sqrt{3}}{36}\kappa\tau^2, \\ \operatorname{Var}_r(g_r) &\leq \min\{\kappa\tau^2/16,\, 1/4\}, \qquad |g_r - \mathbb{E}_r g_r| \leq 1. \end{aligned} \tag{21}$$

The first-order Taylor term vanishes by itemwise logit-bias centering. The remaining estimates only need $|\hat{Y}_A - \mu_B| \leq 1$; no differentiability or margin property of the threshold rule is used.

Write $U = U_{\tau,\gamma}(|D_2|)$. Combining the original Bernstein event with the calibration coverage event gives, with probability at least $1 - \alpha_g - \gamma + o(1)$,

$$|g_{\hat{R}}| \leq \min\left\{1,\ \frac{\sqrt{3}}{36}\kappa U^2 + \sqrt{\frac{\min\{\kappa U^2, 4\}\log(2/\alpha_g)}{8K}} + \frac{\log(2/\alpha_g)}{3K}\right\}. \tag{22}$$

6. Task 2: revised statistic and certification

On D_1, let $m = |D_1|$ and compute the revised paired statistic

$$\begin{aligned}\hat{\Delta}_{\hat{R}} &= \frac{1}{m}\sum_{j\in D_1}\left[\{\hat{Y}_A(x_j)-\hat{\mu}_R(x_j)\}^2-\{y_{B,j}-\hat{\mu}_R(x_j)\}^2\right]\\ &= \frac{1}{m}\sum_{j\in D_1}\{\hat{Y}_A(x_j)-y_{B,j}\}\{1-2\hat{\mu}_R(x_j)\}.\end{aligned} \tag{23}$$

Estimate its standard error by the sample standard deviation of these paired summands divided by √m. The linear and squared-loss implementations must agree, up to numerical rounding.

$$H_0 : \Delta_{\hat{R}}(\Delta_{\hat{R}} + 2g_{\hat{R}}) \le 0, \qquad H_1 : \Delta_{\hat{R}}(\Delta_{\hat{R}} + 2g_{\hat{R}}) > 0. \tag{24}$$

Using the same error-budget allocation and U from page 6, certify when

$$\begin{aligned}&|\hat{\Delta}_{\hat{R}}| - t_{m-1,\,1-\alpha_\Delta/2}\,\widehat{\mathrm{se}}(\hat{\Delta}_{\hat{R}})\\ &> 2\min\left\{1,\ \frac{\sqrt{3}}{36}\kappa U^2 + \sqrt{\frac{\min\{\kappa U^2, 4\}\log(2/\alpha_g)}{8K}} + \frac{\log(2/\alpha_g)}{3K}\right\}.\end{aligned} \tag{25}$$

Thus [S2, Eq. (7)] keeps its right-hand side. Only the main statistic and its paired standard error change. Certification concerns the same strictly nonzero sign; a zero latent or proxy comparison is included in the null.

False-certification guarantee and proof

Retain the logit model, independent reference sampling, the revised moment condition (12), calibration coverage, and the paired CLT with a consistent nondegenerate standard error. For fixed K, over the reference-panel and evaluation/calibration randomness,

$$\limsup_{|D_1|,|D_2|\to\infty} P\big(\text{certify and } \Delta_{\hat{R}}\Delta^* \le 0\big) \le \alpha_\Delta + \alpha_g + \gamma. \tag{26}$$

On the paired confidence event, the left side of (25) is at most $|\Delta_R|$. If the proxy and latent comparisons do not have the same strictly nonzero sign, (11) implies $|\Delta_R| \le 2|g_R|$. On the Bernstein and calibration events, this is at most the right side of (25). A false certification therefore requires one of the three events to fail. The union bound gives (26); mutual independence of those events is unnecessary.

Equation (26) states the joint false-certification probability. It should not be restated as a guarantee conditional on the random panel already giving a misleading comparison.

Implementation order

1. Train probability models and freeze A's validation-selected or prespecified threshold.
2. On D_2, calibrate τ from unthresholded reference probabilities.
3. Specify κ under (17).
4. On D_1, form binary model outputs, the mean reference probability, (23), and its paired standard error.
5. Apply (25) and record the certified direction.

7. Manuscript and experiment edit map

Location	Required edit
[S1] Section 4 Bernoulli Task 1	Replace the compared model probability by its binary decision. Use (3)-(6); retain observed A and C references and the paired test.
[S1] Section 4.1 Population explanation	For the Bernoulli version, replace the additive-bias/variance expression by (7)-(9). Restrict the nonnegative population statement to a common fixed threshold.
[S2] Section 1 Training and outputs	Keep BCE and the probability targets. Add the decision rule (2), threshold selection, and evaluation conditioning.
[S2] Section 2 Targets and errors	Replace the model output in the loss comparisons and distortion. Use (10)-(14); update the probability-scale orthogonality condition.
[S2] Sections 3-4 Bound and calibration	Use the revised contribution in (14). Retain the constants, three logit assumptions, probability outputs, and calibration regularity conditions.
[S2] Sections 5-6 Test and procedure	Replace the main statistic and standard error by (23); retain the right side of the rule. State the false-certification event as in (26).
[S3] Three assumptions	Keep all three displayed logit assumptions. Add or update the separate probability-scale error condition; logit centering does not substitute for it.

Experiment changes relative to probability-output comparisons

Component	Action
Training and data generation	Keep Bernoulli/logit generation and BCE probability fitting. If saved out-of-sample probabilities exist, recompute evaluation from them; thresholding alone does not require retraining.
Threshold	Use 1/2 or a frozen independent-validation choice. Save the criterion, selected value, and tie rule. Never tune against the final test or the same Task 2 reference panel.
Task 1 evaluation	Recompute error differences, S, standard errors, p-values, and empirical rejection rates using binary model outputs.
Task 2 evaluation	Keep reference probabilities and latent probabilities continuous. Recompute the paired statistic, standard error, latent comparison direction, and certification decisions.
Synthetic oracle and summaries	Evaluate the revised latent target. With known probabilities, use $E_X[(\hat{Y}_A - \mu_B)(1 - 2\mu^*)]$ for Δ^*. Report certification rate, correctness among certifications, and joint false-certification rate separately.

Final consistency check. Only A's compared output is thresholded. Task 1 references remain observed binary labels; Task 2 references remain probability functions. Probability fitting and logit calibration retain their original targets. Recompute every reported comparison from the revised statistic, and do not carry over the old Task 1 variance formula or its strict-positivity claim.